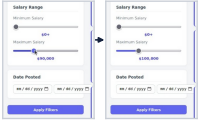
privileged inputs, never part of a trajectory



Summarize
LLM: $H_m \to G_m$

human demos
9 spans H_m, e.g. JobQuest

guide G_m *site-agnostic*

1. locate the range slider
2. drag until its label shows the target
3. click Apply (or equivalent)

pitfall forgetting Apply
done results or URL update

site map
slider 0–2000

dry runs
probe ?max_price

G_m

① Sample
(w, c, x, s_0)

② Suggest
$g,\ \kappa$

③ Execute
$\tau = (o_t, a_t)_{t \le T}$

④ Gate
$\kappa(s_0) = 0,\ \kappa(s_T) = 1$



w ShopHub
c Max Price slider
x 1650
s_0 fresh page

g *"Adjust the Max Price: Any slider to 1650."*

κ `GET ?max_price ∈ [1640, 1670]`

Planner π_0 (untrained base model)
sees o (URL, elements, screenshot), cannot act; each turn emits $u_n = (\gamma, \iota)$ or done(α)

$\mathcal{D}$: 27.6K steps

pooled adapter Δ_{pool}

Held-out SnapLink task, solved in 20 steps:

"Search your shortened links for "archived" and delete the matching links. Then create a short link for https://facebook.com."

1	nav	Click 'My Links'	3 steps
2	text	Type 'archived' in search	4 steps
3	nav	Open 'Old Blog Post (Archived)'	3 steps
4	select	Click the red 'Delete'	3 steps
5	nav	Click 'Create'	3 steps
6	text	Type 'https://facebook.com'	4 steps

done ✓ verifier: links deleted, link created